

	balanced. Current will flow into the positive terminal of cell C and through the ammeter.
4(a)(i)	The magnetic flux linkage through the coil is $NBA \cos \theta$ where N is the number of turns, B the magnetic flux density, A its area and θ , the angle between coil's normal and magnetic field. As coil rotates, the angle θ changes in a sinusoidal manner. When θ is zero, magnetic flux is at maximum and at 90 degrees, the flux is at zero. According to Faraday's law, emf is proportional to the rate of change of magnetic flux linkage. Therefore as change in magnetic flux linkage is sinusoidal, emf induced is also sinusoidal but with a 90 degree phase difference between flux and emf.
4(a)(ii) 1.	Maximum Emf = $3.4 \text{ cm} \times 0.005 \text{ V cm}^{-1} = 0.17 \text{ V}$
4(a)(ii) 2.	Period = $T = 5 \text{ cm} \times 8.0 \times 10^{-3} \text{ s cm}^{-1} = 0.040 \text{ s}$ Frequency = $1/T = 25 \text{ Hz}$
4(b)	Maximum emf, $E_{\text{max}} = NBA\omega$ Magnetic flux density $B = \frac{E_{\text{MAX}}}{NA2\pi f} = \frac{0.17}{120 \times 1.3 \times 10^{-3} \times 2\pi \times 25} = 6.9 \times 10^{-3} \text{ T}$
5(a)	Gravitational potential at a point in a gravitational field is the work done per unit